

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 7
23 February – 1 March 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



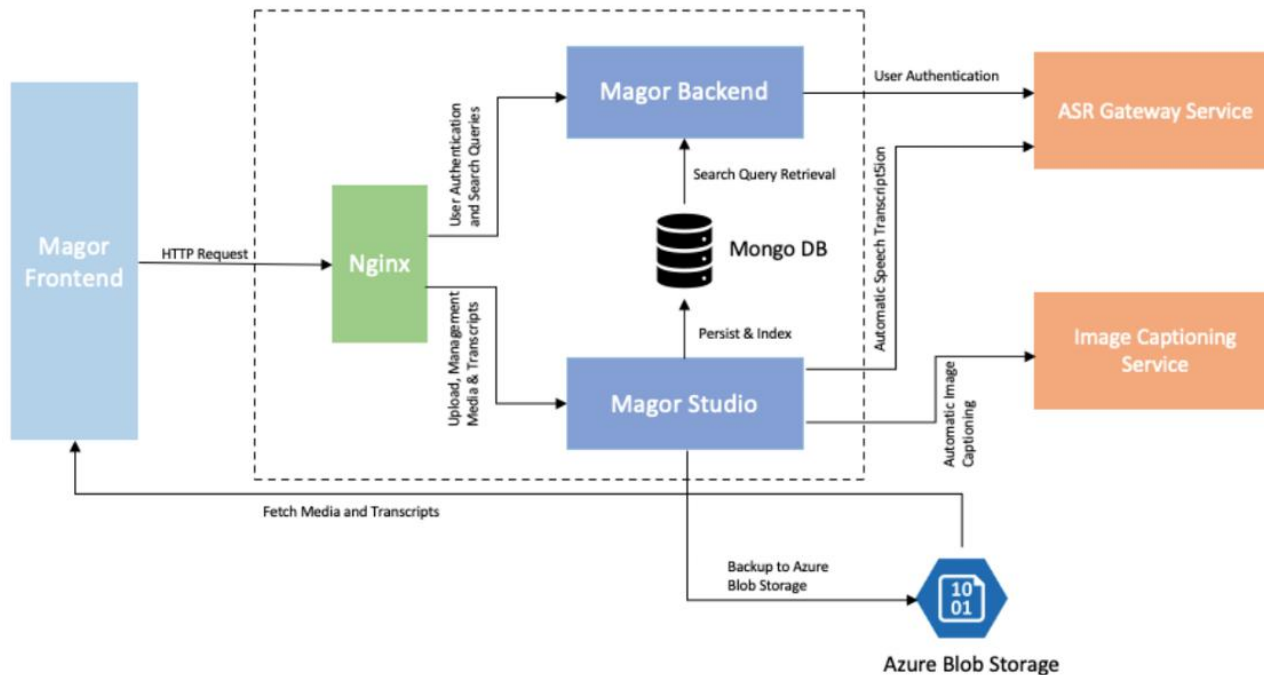
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Fixed dropdown navigation closing when video is playing
2. Implemented ZIP compression for multiple transcript download
3. Updated project dependencies in package.json:
 - Docx
 - Jszip



Task 1 (Dropdown navigation issues)

- Dropdown navigation to other transcript pages would close if video is playing
- Reason: The component was repeatedly re-rendered inside the video player

The screenshot displays a web application interface. At the top, a blue header bar contains the text "Magor" on the left, a search bar with the placeholder "Search" in the center, and a dropdown menu labeled "This recording" with a magnifying glass icon on the right. Below the header, the main content area is divided into two sections. On the left is a video player showing a live stream of a parliamentary session. The video player has a control bar at the bottom with the text "0 matches", "Nothing is playing right now.", and a progress bar showing "0:00 / 2:41". On the right is a transcript section titled "Parliament passes motion considering Pri Media" with an "Edit" button. Below the title are tags: "parliament". Underneath the tags are "Top Words" listed in a horizontal scrollable area: "parliament", "pritam", "integrity", "singh", "privileges", "singapore", "parliamentary", "political", "system", "notes", "public", "high", "court", "singhs". Below the "Top Words" is a "Transcript" section with a "Show options" button. The transcript contains three entries, each starting with "channel1_S1":
1. "The question is that this parliament One. affirms that honesty and integrity are fundamental pillars of Singapore's parliamentary and political system."
2. "So Notes: The judgment in Pritam Singh vs Public Prosecutor (2025). SGHC 242. Where the High Court upheld mr Pritam Singh's conviction. for lying to the committee of privileges. and the findings of the committee of privileges that he had guided. Miss Raizah khan to continue with her lie to parliament."
3. "3. expresses regret at the conduct of mister Pritam Singh. which was dishonorable and unbecoming of a member of parliament. 不错"

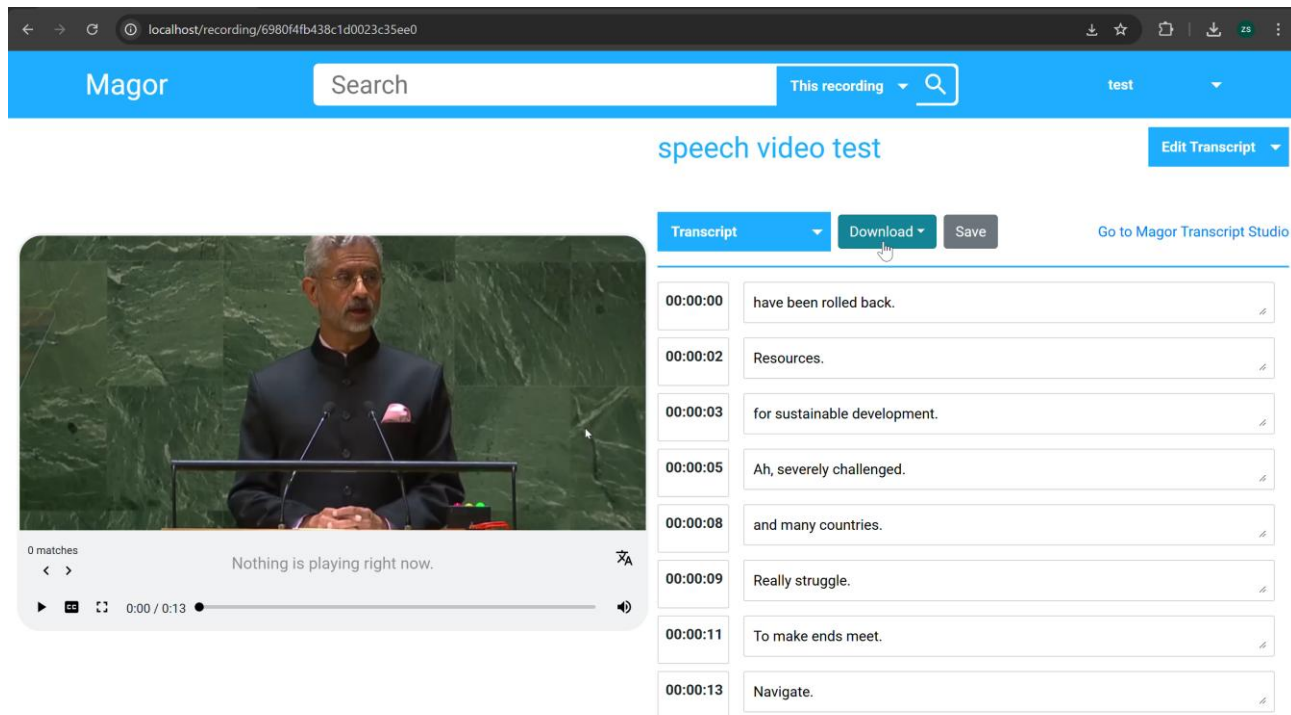
Task 1 (Dropdown navigation issues)

- Fix: Moved the dropdown component outside the frequently re-rendered parent to prevent remounting

The screenshot displays a web application interface for 'Magor'. At the top, there is a blue header bar containing the 'Magor' logo, a search bar, and a dropdown menu currently set to 'This recording'. Below the header, the main content area features a video player on the left and a transcript on the right. The video player shows a live session of the Singapore Parliament, with a speaker at a podium and an audience seated in a large hall. The transcript on the right is titled 'Parliament passes motion considering Pri Media' and includes tags such as 'parliament', 'parliamentary', 'political', 'system', 'notes', 'public', 'high', 'court', 'singhs', 'pritam', 'integrity', 'singh', 'privileges', 'singapore', and 'parliamentary'. The transcript is organized into sections labeled 'channel1_S1', each containing a paragraph of text. A speaker icon is visible in the bottom right corner of the interface.

Task 2 (Multi-Download ZIP Compression)

- Previously implemented:



The screenshot displays the Magor web application interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and a dropdown menu set to 'This recording'. Below the header, the page title 'speech video test' is visible, along with an 'Edit Transcript' button. The main content area features a video player on the left and a transcript list on the right. The video player shows a man speaking at a podium, with a progress bar at the bottom indicating '0:00 / 0:13'. The transcript list on the right contains eight entries, each with a timestamp and a text snippet. A 'Download' button is highlighted in the top right of the transcript section.

Magor Search This recording test Edit Transcript

speech video test

Transcript Download Save Go to Magor Transcript Studio

00:00:00 have been rolled back.

00:00:02 Resources.

00:00:03 for sustainable development.

00:00:05 Ah, severely challenged.

00:00:08 and many countries.

00:00:09 Really struggle.

00:00:11 To make ends meet.

00:00:13 Navigate.



Task 2 (Multi-Download ZIP Compression)

- ZIP compression Demo:

The screenshot displays the Magor web interface. At the top, there is a blue header bar with the 'Magor' logo, a search bar, and a dropdown menu labeled 'This recording'. Below the header, the text 'speech video test' is visible, along with an 'Edit Transcript' button. The main content area features a video player on the left and a transcript list on the right. The video player shows a man speaking at a podium, with a progress bar at the bottom indicating '0:00 / 0:13'. The transcript list on the right contains several entries with timestamps and text, each with a small icon to its right. A 'Download' button is highlighted with a mouse cursor.

Magor Search This recording test

speech video test Edit Transcript

Transcript Download Save Go to Magor Transcript Studio

00:00:00 have been rolled back.

00:00:02 Resources.

00:00:03 for sustainable development.

00:00:05 Ah, severely challenged.

00:00:08 and many countries.

00:00:09 Really struggle.

00:00:11 To make ends meet.

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Task 2 (Multi-Download ZIP Compression)

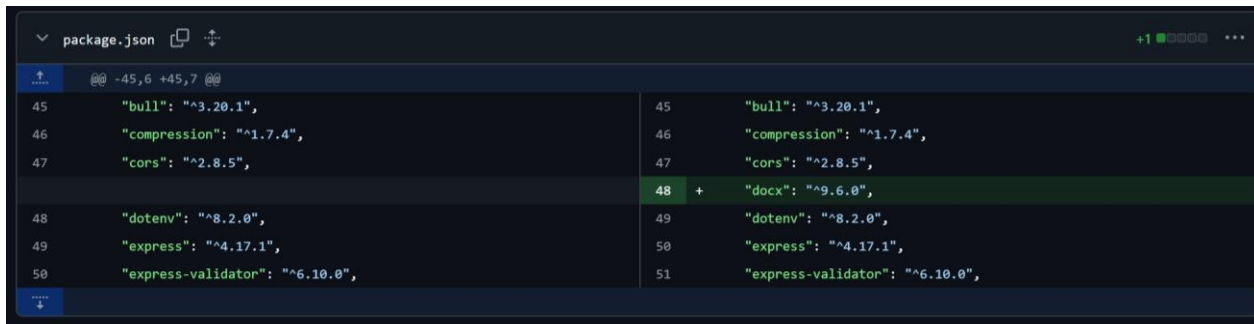
- Current state of transcript downloads:

The screenshot displays the Magor web application interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and navigation links like 'This recording' and 'test'. Below the header, the main content area features a video player on the left and a transcript list on the right. The video player shows a man speaking at a podium, with a progress bar at the bottom indicating '0:00 / 0:13'. The transcript list on the right has a 'Download' button and a 'Save' button. It contains a list of transcript entries with timestamps and text.

Timestamp	Transcript Text
00:00:00	have been rolled back.
00:00:02	Resources.
00:00:03	for sustainable development.
00:00:05	Ah, severely challenged.
00:00:08	and many countries.
00:00:09	Really struggle.
00:00:11	To make ends meet.
00:00:13	Navigate.

Task 3 (Update dependencies in package.json)

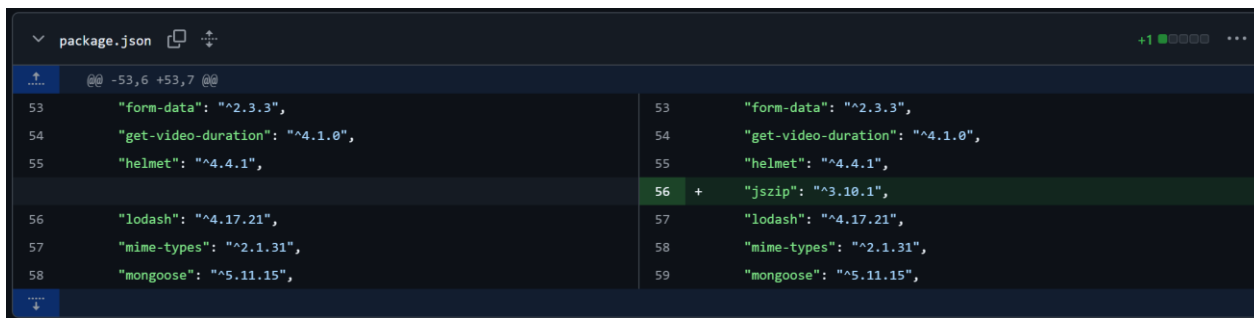
- Previously, additional libraries were only added in `node_modules`
- Updated them in `package.json` to track project dependencies



```
package.json
```

45	"bull": "^3.20.1",	45	"bull": "^3.20.1",
46	"compression": "^1.7.4",	46	"compression": "^1.7.4",
47	"cors": "^2.8.5",	47	"cors": "^2.8.5",
		48	+ "docx": "^9.6.0",
48	"dotenv": "^8.2.0",	49	"dotenv": "^8.2.0",
49	"express": "^4.17.1",	50	"express": "^4.17.1",
50	"express-validator": "^6.10.0",	51	"express-validator": "^6.10.0",

docx



```
package.json
```

53	"form-data": "^2.3.3",	53	"form-data": "^2.3.3",
54	"get-video-duration": "^4.1.0",	54	"get-video-duration": "^4.1.0",
55	"helmet": "^4.4.1",	55	"helmet": "^4.4.1",
		56	+ "jszip": "^3.10.1",
56	"lodash": "^4.17.21",	57	"lodash": "^4.17.21",
57	"mime-types": "^2.1.31",	58	"mime-types": "^2.1.31",
58	"mongoose": "^5.11.15",	59	"mongoose": "^5.11.15",

jszip

Plans for Next Week

- Update Transcript Studio UI
 - Adjust button order (Download after publish)
- Prevent empty tags from appearing in top transcript words
- Restore admin page functionality
 - Recordings Tab
 - Fix upload date display and sorting
 - Fix recording storage path display
 - Edits and deletes to be successfully updated

